Supplementary Methods and Results

Phase-Informed ECMS with Soft SOC Safeguarding for a Virtual Hybrid Agricultural Tractor

Chenyu Cui, Peng Wang and Yucheng Zhang

S1 Defined Component Maps and Numerical Verification

All primary ratings are researcher-defined virtual inputs. None constitutes a measured rating of a target tractor. The engine regression shape is traceable to Devyanin et al. (2023), DOI 10.22314/2073-7599-2023-17-4-68-74. Its source-scale efficiency is transformed rather than used as a calibrated 300 kW map. For requested virtual power P_e, relative load (percent) is

$$M = \min(105, \max(0, 100|P_e|/300000)).$$

At a source speed n (rpm),

$$\begin{aligned} G(n, M) &= -0.7219 + 0.0026123n + 0.00493M + 0.0001743nM \\ &\quad - 0.0000002092n^2 - 0.0000172M^2 \quad [\text{L/h}], \\ P_{src} &= (nM\,592/(100 \cdot 9550))1000 \quad [\text{W}], \\ g_e &= 1000(0.820G)/(P_{src}/1000) \quad [\text{g/kWh}], \\ r_e(n, M) &= \max(0.25, (3600/(42.7g_e))/0.329). \end{aligned}$$

For $P_{src} \leq 100$ W, $r_e = 0.30$ instead. Linear interpolation of r_e uses speed nodes 750, 1100, 1500, 1800, 2150 and 2500 rpm, with endpoint clipping. The implemented efficiency is $\eta_e = \max(0.10, \min(0.42, 0.42r_e))$. The normalizer 0.329, low-power guard and efficiency clipping are model choices. Consequently, even the normalized shape is a guarded proxy. No preservation of absolute source consumption or target-engine torque is claimed.

Both machines use the same prescribed power-only law, with P in W:

$$\eta_m(P) = \max\{0.70, \min[0.96, 0.96(0.75 + 0.25e^{-((|P|-45000)/50000)^2})]\}.$$

Although the software interface accepts speed, speed does not affect this efficiency. Machine power and speed are independently screened, but no low-speed torque envelope is supplied. Main-text equations specify the ideal planetary split and the OCV–resistance battery. The 200 kWh, 650 V input gives $Q = 307.6923$ Ah. At target SOC, +180 kW requires +282.446 A and has 3.590 kW ohmic loss; −180 kW requires −271.808 A and has 3.325 kW loss. These are algebraic outputs, not pack safety validation.

A reproducible 1200-state/action-input sampling check (seed 5240) enumerated 42084 execution candidates; 28093 passed the instantaneous physical screens. Maximum absolute residuals were 5.83×10^{-11} W for mechanical balance, 7.43×10^{-10} W for battery power and zero at reported floating-point precision for kinematic, bus-definition and SOC-update identities. Rejections were 3930 MG1-power, 7561 MG2-power and 2500 battery-bus cases. The check verifies numerical consistency of these equations; it cannot establish physical representativeness or validate the imposed envelopes. The raw ledger is `model_consistency_checks.json`.

An additional applied-command audit covers all 20 independent paths for the adaptive, shared-safeguard fixed and matched-baseline arms: 60 trajectories, 36000 samples, 397 starts and 337 stops. It finds no violations of the 60 kW cold-start point, 40 kW/s running ramp, five-second minimum

on-time, one-second minimum off-time, bus limit or physical SOC bounds. All 60 meet the strict terminal tolerance. Default and explicitly enabled module settings give identical non-timing outputs; the first submitted seed retains exact fuel parity. These execution checks do not prove recursive feasibility or validate hardware.

The alternative map family is an explicitly assumed efficiency grid. Speed nodes are 1000, 1500, 2000 and 2400 rpm; relative-load nodes are 0, .2, .4, .6, .8 and 1. Its unmodified rows are

$$\begin{pmatrix} .18 & .27 & .35 & .40 & .40 & .37 \\ .20 & .31 & .39 & .43 & .42 & .39 \\ .19 & .30 & .38 & .41 & .40 & .37 \\ .17 & .27 & .35 & .38 & .37 & .34 \end{pmatrix}.$$

For low-load input ℓ, every first entry becomes ℓ and every second entry becomes $\max(\ell, \min(.43, \ell + \eta_{base,2} - .25))$. Remaining entries stay fixed. Bilinear interpolation uses $|P_e|/300000$ clipped to $[0, 1]$, endpoint speed clipping and output limits $[.12, .43]$. Thus the low-load perturbations also use a different peak cap from the primary 0.42 map; this is a map-family sensitivity, not identification of one isolated physical parameter.

S2 Public Reference, Safeguard and Baseline

Let $\tau = (k\Delta t) \bmod 270$, $s_* = .55$ and $T_r = (N - k)\Delta t$. The raw phase offset is

$$\delta(\tau) = \begin{cases} .015, & 0 \le \tau < 120, \\ .015 - .020(\tau - 120)/120, & 120 \le \tau < 240, \\ -.005 + .005(\tau - 240)/30, & 240 \le \tau < 270. \end{cases}$$

The public reference is $s_k^{ref} = s_* + a(T_r)\delta(\tau)$, where $a = 1$ for $T_r \ge 270$, $a = (T_r - 120)/150$ for $120 < T_r < 270$, and $a = 0$ otherwise. This schedule is specified by the researcher, not an optimized future-dependent reference. It remains the same in low-load and other cycle audits, exposing its task dependence.

The economic proposal is minimized over currently hard-feasible candidates after the conditional engine-on reserve preference. The successor filter uses $|s_{k+1} - s_k^{ref}| \le .045[(N - k - 1)/(N - 1)]^2$. If that soft set is empty, the selected hard-feasible candidate minimizes successor SOC distance, with equivalent cost as the secondary criterion. If the hard set is empty, simulation stops and the failure is retained. The soft corridor is not an invariant set, a reachable set or a robust safety certificate. At the last step its width is zero, while final acceptance permits 10^{-4} SOC error; no appended charge-restoration action is used.

The matched load-following baseline also uses the public reference, the same hard candidate set and conditional reserve. Its target bus power is

$$P_b^* = \mathrm{clip}_{[-P_{max}, P_{max}]}\left(\frac{(s_k - s_k^{ref})E_{Wh}3600}{\max(\Delta t, T_r)}\right).$$

It restricts successor SOC to a band $.1\epsilon + .002\max(0, (N - k - 1)/(N - 1))$, $\epsilon = 10^{-4}$, if candidates exist; otherwise it uses the nearest-SOC candidate. Selection minimizes $|P_b - P_b^*|$, then instantaneous fuel, then change in engine power. This band and its fallback are part of the baseline, not the ECMS safeguard, and the baseline is explicitly stronger than unregulated engine-only load following. It is evaluated for terminal validity independently of ECMS.

Conventional fixed ECMS uses a constant .55 reference, constant equivalence factor, the common execution constraints and reserve, but no successor-SOC filter. Conventional adaptive ECMS has proportional SOC feedback to .55 and the same restrictions, again without the filter. Fixed ECMS

with the shared safeguard instead receives precisely the same phase, reserve, SOC filter and execution constraints as the primary adaptive arm, with $K_p = 0$. The latter is the decisive same-mechanism comparator.

S3 Task Generator and Seed Separation

The nominal cycle is a sequence of idle (30 s, 0 m/s, 12000 N draft, zero grade), transport (90 s, 2.2 m/s, 45000 N, zero grade), ploughing (120 s, 1.5 m/s, 145000 N, grade .05), and turn (30 s, .8 m/s, 30000 N, zero grade). The sequence repeats until 600 samples are obtained. Inputs interpolate toward the next segment in each segment's last 5 s; draft is floored at 10000 N. The low-load sequence changes these four durations to 60, 120, 60 and 30 s. The abrupt high-variability profile is separately defined in `code/common/profiles.py` and uses 300 s in the revised audit. It is intentionally not smoothed into primary feasibility.

Stochastic disturbances use Python's seeded pseudorandom generator. Every 100 samples, the soil multiplier target changes by a Gaussian increment with standard deviation .06 and is clipped to $[.82, 1.28]$. The grade-bias target changes by a Gaussian increment with standard deviation .004 and is clipped to $[-.025, .025]$. Within each block the current values move linearly from the block-start values to the targets with fraction $((k \bmod 100) + 1)/100$. At the revised 1 s sample interval these blocks are 100 s, rather than the 10 s blocks of the legacy 0.1 s protocol. There is no retrospective disturbance smoothing or controller-dependent task generation.

Purpose	Seeds	Controller tuning allowed?
Submitted development / replay	3100–3105 / 3120–3139	Original selection / none
New conventional development	5100–5105	Equal 15-candidate budgets
Independent primary and ablation	5120–5139	No
Additional stochastic cycle	5160–5179	No
Action-grid audit	5180–5185	No
Map and capacity audit	5200–5219	No
Numerical model check	5240	Not a tuning experiment

Disjoint seeds prevent reuse of those generated paths for parameter selection. They do not create independence from the shared cycle design or virtual map. Seed 5120 is the declared illustration because it is the first independent identifier; its published trajectory summary is retained in the article.

S4 Parameter Selection and Module Protocol

The original 75-candidate search used the Cartesian product

$$\lambda_0 \in \{.8, 1, 1.2\}, \qquad K_p \in \{20, 40, 60, 80, 120\}, \qquad b_0 \in \{.008, .015, .025, .035, .045\},$$

with a development SOC tolerance of .001. It retained six-valid-path candidates, preferred those with mean interventions at most 100 and then maximized mean saving, breaking ties by fewer interventions and smaller maximum SOC error. The selected $(1, 40, .045)$ was frozen. A subsequent strict 10^{-4} audit on seeds 3120–3139 is the source of the submitted 5.676% normalized result. Development success at .001 is not described as development certification at 10^{-4}. Its full original ledger is retained in `submitted_reference.json`.

The submitted load-following controller's internal band was computed with its default .001 setting, although its realized terminal errors all satisfy 10^{-4}. The new matched baseline uses the explicit 10^{-4} setting in S2. This baseline-band change can slightly change the fuel denominator; it is not

a retuning of ECMS. For exact attribution to the submitted 5.676%, the submitted-path ablation below uses the original recorded baseline. New independent comparisons instead use the revised stricter-band baseline for every arm. We do not interpret differences between these protocols as improvement over the submitted saving.

The new fixed-factor development explores $.3, .4, \ldots, 1.7$. Adaptive development explores $\lambda_0 \in \{.8, 1, 1.2\}$ and $K_p \in \{20, 40, 60, 80, 120\}$. Each has 15 candidates on six new development seeds, with 10^{-4} terminal acceptance. The shared-safeguard fixed arm also uses those 15 factors. Ranking maximizes terminal-valid count, then completed count, then completed-only normalized diagnostic saving, and finally minimizes maximum SOC error. The diagnostic tie breaker does not certify invalid paths. All 45 candidates and their individual paths are retained. They select $\lambda = 1$ for both fixed families and $(1, 60)$ for conventional adaptive ECMS. Held-out results do not enter selection.

All primary ablations use frozen $(1, 40, .045)$ except $K_p = 0$ when adaptation is deleted. One-at-a-time removals disable phase, SOC filter or reserve. The stepwise sequence is (1) constant-reference adaptive ECMS without reserve/filter, (2) add public phase, (3) add conditional reserve, and (4) add the SOC filter. Step 3 equals the one-at-a-time no-filter arm; Step 4 equals the complete arm. The primary cold-start and dwell rules remain common physical execution restrictions. Their removal would change actuator authority, so no benefit is attributed to deleting them. The analysis establishes module dependence under these fixed restrictions, not a factorial decomposition of all mechanisms or a claim that the actuator restrictions themselves improve economy.

S5 Complete Comparison and Attribution Results

The following post-review ablation uses submitted seeds 3120–3139 and the exact submitted baseline; it was not used for parameter selection. All 20 paths complete in every arm.

Submitted-path variant	Valid pairs	Saving (%)	Max $\lvert\Delta s\rvert$	Corrections
Submitted full adaptive	20/20	5.676	5.52e-05	93.5
Without adaptation	19/20	7.079	0.00245	148.9
Without phase	17/20	8.041	0.00712	59.4
Without SOC filter	0/20	–	0.00494	0.0
Without reserve	20/20	5.685	5.52e-05	91.5

The independent results below test the same configuration and module interpretation on disjoint seeds with the new common baseline. The tables below use completed, terminal-valid same-seed baseline/controller intersections for savings and paired Student-t intervals. Counts always retain all 20 attempted paths. Intervals for very small surviving subsets are correspondingly wide; no inference from two paths is promoted to a general comparison.

Variant	Complete	Valid pairs	Saving (%)	95% interval
Frozen adaptive + safeguard	20/20	20/20	5.951	[5.352, 6.549]
Fixed + same safeguard	20/20	20/20	7.091	[6.025, 8.157]
No public phase	20/20	16/20	8.786	[7.312, 10.261]
No SOC safeguard	20/20	0/20	–	–
No reserve	20/20	20/20	5.958	[5.342, 6.573]
Step 1: adaptive, constant reference	20/20	0/20	–	–
Step 2: add public phase	20/20	0/20	–	–
Developed conventional fixed	20/20	0/20	–	–
Developed conventional adaptive	20/20	2/20	10.247	[-1.710, 22.205]
Developed fixed + same safeguard	20/20	20/20	7.091	[6.025, 8.157]
Matched load following	20/20	20/20	0.000	[0.000, 0.000]

Completed-only diagnostic saving in the following table includes SOC-invalid trajectories. It is retained to expose the fuel/charge confound and is not a successful charge-sustaining saving. SOC maxima and intervention/start means likewise refer to all completed paths.

Variant	Diagnostic (%)	Max $\|\Delta s\|$	Corrections	Starts
Frozen adaptive + safeguard	5.951	1.88e-06	94.8	7.20
Fixed + same safeguard	7.091	2.07e-06	150.5	6.90
No public phase	8.117	0.00432	60.0	5.20
No SOC safeguard	8.368	0.00627	0.0	3.00
No reserve	5.958	1.69e-06	93.0	7.65
Step 1: adaptive, constant reference	8.392	0.00961	0.0	2.80
Step 2: add public phase	8.193	0.00693	0.0	2.65
Developed conventional fixed	10.337	0.0218	0.0	3.00
Developed conventional adaptive	8.599	0.00528	0.0	2.90
Developed fixed + same safeguard	7.091	2.07e-06	150.5	6.90
Matched load following	0.000	1.63e-06	0.0	5.75

The adaptive-minus-shared-fixed paired saving difference is -1.140 percentage points (95% interval $[-1.660, -.620]$, 20 pairs), as recorded in `revision_provenance.json`. Adaptation yields 37.0% fewer mean proposal corrections but does not lower mean starts (7.2 versus 6.9). This is a fuel–intervention trade-off, not evidence of superior fuel economy or lower mechanical wear. Reserve deletion changes mean saving by only about .007 percentage points on this set; it does not demonstrate tolerance to arbitrary demand jumps. Removing phase increases terminal-invalid paths from zero to four. Removing the SOC filter leaves all 20 paths completed but none terminal-valid. Conditional valid-subset percentages must be read with those counts.

S6 Same-Gain Audits and Failure Ledger

Every row below uses the frozen $K_p = 40$, $\lambda_0 = 1$, $b_0 = .045$ and strict 10^{-4} tolerance. Previously submitted V2 cross-cycle/map audits with $K_p = 60$ are historical and are not evidence for this exact frozen configuration. "Hard fail" counts controller trajectories terminated by an empty hard set; completed but SOC-invalid paths are additionally visible through validity counts. The published tables retain the aggregate baseline and controller records.

Audit	Baseline valid	ECMS valid	Hard fail	Pairs	Saving (%)	95% interval
deterministic	1/1	1/1	0	1/1	6.604	–
low_load	1/1	1/1	0	1/1	9.871	–
high_variability	0/1	0/1	1	0/1	–	–
new_stochastic	20/20	20/20	0	20/20	6.144	[5.540, 6.748]
grid_10000	6/6	6/6	0	6/6	6.168	[4.817, 7.519]
grid_5000	6/6	6/6	0	6/6	6.115	[4.879, 7.351]
grid_2500	6/6	6/6	0	6/6	5.885	[4.609, 7.160]
map_literature	19/20	20/20	0	19/20	5.724	[4.975, 6.473]
map_0.15	19/20	18/20	1	18/20	8.011	[6.723, 9.300]
map_0.25	19/20	18/20	1	18/20	6.509	[5.400, 7.619]
map_0.35	19/20	18/20	1	18/20	6.086	[4.962, 7.210]
capacity_120000	18/20	18/20	0	18/20	4.240	[3.708, 4.772]
capacity_200000	19/20	20/20	0	19/20	5.724	[4.975, 6.473]
capacity_280000	20/20	20/20	0	20/20	6.204	[5.416, 6.992]

The failure ledger is summarized below. Failure strings report the realized demand, engine power and dwell state at the empty-action step. No incomplete trajectory contributes a fuel saving, and completed-only averages cannot hide the following cases.

- high_variability, baseline, seed 0: no feasible power-split action (step=159, demand_w=249442.4, engine_w=80000.0, on_s=159.0, off_s=0.0)

- high_variability, controller, seed 0: no feasible power-split action (step=159, demand_w=249442.4, engine_w=85000.0, on_s=159.0, off_s=0.0)
- map_literature, baseline, seed 5204: terminal SOC error -0.000974926.
- map_0.15, baseline, seed 5204: terminal SOC error -0.000974926.
- map_0.15, controller, seed 5204: no feasible power-split action (step=390, demand_w=335363.7, engine_w=240000.0, on_s=360.0, off_s=0.0)
- map_0.15, controller, seed 5206: terminal SOC error -0.00244865.
- map_0.25, baseline, seed 5204: terminal SOC error -0.000974926.
- map_0.25, controller, seed 5204: no feasible power-split action (step=390, demand_w=335363.7, engine_w=240000.0, on_s=360.0, off_s=0.0)
- map_0.25, controller, seed 5206: terminal SOC error -0.0024478.
- map_0.35, baseline, seed 5204: terminal SOC error -0.000974926.
- map_0.35, controller, seed 5204: no feasible power-split action (step=390, demand_w=335363.7, engine_w=240000.0, on_s=361.0, off_s=0.0)
- map_0.35, controller, seed 5206: terminal SOC error -0.00246308.
- capacity_120000, baseline, seed 5204: terminal SOC error -0.00820802.
- capacity_120000, baseline, seed 5206: terminal SOC error -0.00560805.
- capacity_120000, controller, seed 5204: terminal SOC error -0.00473383.
- capacity_120000, controller, seed 5206: terminal SOC error -0.00215451.
- capacity_200000, baseline, seed 5204: terminal SOC error -0.000974926.

Grid tests change the engine-power grid only. Capacity tests change energy capacity and thus the derived ampere-hour capacity, leaving voltage and resistance fixed; they are controlled mathematical perturbations, not redesigned physically equivalent packs. The 0.15, 0.25 and 0.35 map inputs refer to the assumed grid law in S1, not measured low-load efficiencies. These finite checks do not establish robustness to joint map, thermal, task-boundary and demand uncertainty.

S7 Historical Mechanism and Full-Information Records

This section preserves earlier submitted findings while removing them from the revised same-platform headline comparison. The legacy model uses a different plant implementation, 150 kW electrical limits, 0.1 s task sampling and its own operating-line baseline. Its MPC uses predictive constraints and a backup construction separate from the new single-successor SOC safeguard. Historical certification labels refer only to those recorded finite executions and their defined checks; they must not be read as proof that the revised safeguard is recursively feasible.

Historical condition	Controller / statistic	Saving (%)
Literature-shaped frozen framework	Map-aware ECMS	−.251
Literature-shaped frozen framework	Terminal-set MPC	−.704
Steep-island frozen framework	Map-aware ECMS	10.491
Steep-island frozen framework	Terminal-set MPC	−3.219
Literature-shaped mechanism holdout	History-only MPC, $H = 8$	−.168
Steep-island mechanism holdout	History-only MPC, $H = 8$	−2.079
Favorable full-information replay	Raw direct-fuel reduction	21.611
Favorable full-information replay	Terminal-energy adjusted reduction	19.828

Negative numbers denote more fuel than the corresponding legacy baseline. Storage (120–280 kWh) and causal predictor-gain (0, .5, 1) sweeps also retained negative MPC savings. Legacy map sweeps can change the ECMS sign and magnitude; therefore 10.491% is a mechanism observation, not replication of the revised 5.676% primary claim. The published supplementary tables retain the historical framework, map, mechanism, storage/predictor, demand-boundary and oracle summaries. They are identified as retained records rather than newly rerun experiments.

The favorable full-information task uses a deliberately steep map, complete nominal future, 150 kW electrical limits, 200 kWh energy capacity and a ±.01 terminal SOC band. Its replay ends at SOC error +.00669; the legacy baseline ends at +.00950, leaving the oracle .563 kWh lower. A .42 conversion-efficiency adjustment adds .136 L to account for that difference and changes raw 21.611% saving to 19.828%, i.e., approximately 20%. This is not the primary 10^{-4} terminal criterion, an online controller, or a field prediction. The separately attempted literature-shaped oracle ends at −.0310 SOC error on continuous replay and is rejected. The former 24.087% grid result was also rejected after replay. Finite-grid diagnostics and closeness of two grid values do not prove global optimality, so "theoretical upper bound" is replaced by "favorable full-information diagnostic."

Reproduction

The public repository retains the executable virtual-plant source, declared parameter manifests, clean manuscript source and this PDF. Frozen revision JSONs, per-sample trajectories, checkpoints and historical audit archives are intentionally excluded. Using Python 3.12 with NumPy, SciPy and Matplotlib, run the minimal source-only check from a fresh checkout:

```
./reproduce_paper2.sh --workspace /tmp/paper2-reproduction
```

The runner creates a short seeded adaptive-ECMS and dynamic-baseline comparison in the external workspace. It does not read historical results and is a source-level theoretical recomputation; it does not claim byte-for-byte parity with omitted frozen outputs, field validation or universal performance.